

Enterprise Infrastructure Planning
for ABC Startup Solutions

ITEP 414 - System Administration and Maintenance
Week 2 Portfolio Project

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PART 1 – COMPANY PROFILE

1.1 Company Name

ABC Startup Solutions

1.2 Nature of Business

ABC Startup Solutions is a newly established software development company. The company currently has 20 employees and operates in a single office floor.

1.3 Company Vision

To establish ABC Startup Solutions as a reliable software development company supported by an organized, secure, and efficient information technology infrastructure.

1.4 Company Mission

ABC Startup Solutions aims to support its software development operations through reliable technology resources, organized IT infrastructure, secure systems, and effective management of information technology resources.

1.5 Office Location

ABC Startup Solutions, Main Office, Poblacion, Liliw, Laguna, Philippines

1.6 Organizational Structure

ABC Startup Solutions is composed of four departments: Information Technology, Human Resources, Finance, and Sales.

The Information Technology Department is responsible for the company's technology-related operations and IT infrastructure. The Human Resources Department handles employee-related functions. The Finance Department manages financial activities and records. The Sales Department handles the company's sales-related activities.

1.7 Employee Distribution

Department	Number of Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

The employee distribution follows the company scenario provided in the project.

PART 2 – ENTERPRISE HARDWARE INVENTORY

2.1 Hardware Inventory

The following hardware inventory is proposed for ABC Startup Solutions based on its 20 employees, four departments, and single-office-floor setup.

Asset ID	Hardware	Quantity	Department	Purpose
HW-001	Desktop Computer	20	All Departments	Provides a primary workstation for each employee.
HW-002	Laptop	5	Information Technology	Provides portable computers for IT-related activities.
HW-003	Server	1	Information Technology	Provides centralized server resources for the company.
HW-004	Router	1	Information Technology	Provides routing and network connectivity.

HW-005	Switch	1	Information Technology	Connects wired devices within the office network.
HW-006	Printer	2	All Departments	Provides shared printing resources for company documents.
HW-007	UPS	3	Information Technology	Provides power protection for critical IT equipment.
HW-008	Wireless Access Point	2	All Departments	Provides wireless network connectivity throughout the office.
HW-009	NAS Storage	1	Information Technology	Provides centralized network storage.
HW-010	External Backup Drive	2	Information Technology	Provides additional storage for backup copies.
HW-011	Monitor	20	All Departments	Provides a display for each desktop workstation.

2.2 Hardware Selection Justification

The proposed hardware inventory is based on the company's 20 employees and single-office-floor environment. Twenty desktop computers are proposed so that each employee has a primary workstation. Five laptops are assigned to the Information Technology Department to provide portable computing resources for IT activities.

One server is proposed as the initial centralized server resource. One router and one switch provide the basic network infrastructure required to connect the company's devices. Two printers provide shared printing resources for the four departments.

Three UPS units are proposed to provide power protection for critical IT equipment. Two wireless access points are proposed to provide wireless connectivity across the office. One NAS provides centralized network storage, while two external backup

drives provide additional backup storage. Finally, 20 monitors are proposed to support the 20 desktop workstations.

PART 3 – ENTERPRISE SOFTWARE INVENTORY

3.1 Software Inventory

The following software is proposed for ABC Startup Solutions based on the software requirements identified in the project.

Software	Version	License	Purpose
Windows 11 Pro	Current supported version	Commercial	Operating system for employee workstations.
Ubuntu Server	LTS version	Open Source	Server operating system for centralized server services.
Microsoft Office	Current supported version	Commercial	Productivity software for documents, spreadsheets, and presentations.
Visual Studio Code	Current version	Free	Code editor for software development.
Git	Current version	Open Source	Version control for software development projects.
GitHub Desktop	Current version	Free	Provides a graphical interface for Git and GitHub repository management.
VirtualBox	Current version	Open Source	Provides virtualization for testing and running virtual machines.

Google Chrome	Current version	Free	Web browser for accessing web-based resources and services.
Microsoft Defender	Included with Windows	Included	Provides security and malware protection for Windows computers.
AnyDesk	Current version	Commercial/Free depending on use	Provides remote access and technical support capabilities.
7-Zip	Current version	Open Source	Compresses and extracts files and archives.

3.2 Software Requirements Explanation

Windows 11 Pro will be used as the operating system for employee workstations. It provides the basic environment required for everyday company activities.

Ubuntu Server will be used as the operating system for the company's server. It provides a server platform for centralized IT services.

Microsoft Office will provide productivity applications for creating and managing documents, spreadsheets, and presentations.

Visual Studio Code will support software development and code editing activities.

Git will provide version control for software development projects and help manage changes to source code.

GitHub Desktop will provide a graphical interface for working with Git and GitHub repositories.

VirtualBox will allow IT personnel to create and manage virtual machines for testing and other system administration activities.

Google Chrome will provide web browsing capabilities for employees and access to web-based services.

Microsoft Defender will provide security and malware protection for Windows workstations.

AnyDesk will provide remote access capabilities that can support technical assistance and system administration.

7-Zip will be used to compress and extract files, which is useful when handling software packages, documents, and archived files.

PART 4 – ENTERPRISE NETWORK INVENTORY

4.1 Network Equipment Inventory

The following networking equipment is proposed for ABC Startup Solutions based on its 20 employees and single-office-floor environment.

Asset ID	Network Equipment	Quantity	Purpose
NET-001	ISP Modem	1	Provides the connection between the office network and the Internet Service Provider.
NET-002	Router	1	Routes traffic between the internal network and external networks.
NET-003	Firewall	1	Provides network security and controls network traffic.
NET-004	Switch	1	Connects wired computers and network devices.
NET-005	Access Point	2	Provides wireless network access within the office.
NET-006	Patch Panel	1	Organizes and terminates network cable connections.
NET-007	CAT6 Cables	25	Provides wired network connections between devices.

NET-008	RJ45 Connectors	50	Used to terminate CAT6 network cables.
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4.2 Network Inventory Justification

One ISP modem is proposed to provide Internet connectivity to the office. One router will manage routing between the internal network and external networks, while one firewall will provide an additional layer of network security.

One switch will provide wired connectivity for the company's computers and other network devices. Two access points are proposed to provide wireless connectivity throughout the single office floor.

One patch panel will be used to organize the network cabling. Twenty-five CAT6 cables are proposed to provide enough initial wired connections for employee workstations and network devices, with additional cables available for infrastructure connections and limited expansion. Fifty RJ45 connectors are proposed to provide connectors for cable installation and replacement needs.

PART 6 – SYSTEM ADMINISTRATION ROLES

6.1 Helpdesk Technician

Responsibilities

A Helpdesk Technician provides first-level technical support to users. Responsibilities include receiving and resolving technical issues, assisting users with hardware and software problems, troubleshooting common computer issues, documenting support requests, and escalating problems that require higher-level technical assistance.

Skills

Important skills include troubleshooting, problem-solving, communication, customer service, basic hardware and software knowledge, operating system knowledge, and documentation skills.

Common Tools

Common tools may include ticketing systems, remote support software, operating system diagnostic tools, hardware diagnostic utilities, and communication tools.

Certifications

Relevant certifications may include CompTIA A+ and other entry-level IT support certifications.

6.2 Network Administrator

Responsibilities

A Network Administrator manages and maintains an organization's network infrastructure. Responsibilities include configuring network devices, monitoring network performance, troubleshooting connectivity problems, maintaining network security, managing network access, and documenting network configurations.

Skills

Important skills include networking fundamentals, TCP/IP, network troubleshooting, network security, device configuration, problem-solving, and technical documentation.

Common Tools

Common tools include routers, switches, firewalls, network monitoring tools, diagnostic utilities, and network configuration tools.

Certifications

Relevant certifications may include Cisco Certified Network Associate (CCNA) and CompTIA Network+.

6.3 Linux System Administrator

Responsibilities

A Linux System Administrator manages Linux-based systems and servers. Responsibilities include installing and configuring Linux systems, managing users and

permissions, monitoring system performance, maintaining system security, managing software packages, troubleshooting problems, and performing backups.

Skills

Important skills include Linux command-line administration, file and permission management, system troubleshooting, networking, security, shell scripting, and server management.

Common Tools

Common tools include the Linux command line, package management tools, system monitoring utilities, SSH, text editors, and shell scripting tools.

Certifications

Relevant certifications may include Linux Foundation certifications and Red Hat certifications.

6.4 Cloud Administrator

Responsibilities

A Cloud Administrator manages an organization's cloud-based infrastructure and services. Responsibilities may include configuring cloud resources, managing user access, monitoring cloud services, maintaining security, managing storage and computing resources, and supporting cloud-based systems.

Skills

Important skills include cloud computing concepts, networking, security, identity and access management, virtualization, monitoring, troubleshooting, and resource management.

Common Tools

Common tools include cloud provider management consoles, command-line interfaces, monitoring tools, identity and access management tools, and cloud resource management utilities.

Certifications

Relevant certifications may include entry-level or associate-level certifications from major cloud service providers.

6.5 Collaboration of the Four Professionals

The four professionals work together to maintain the organization's IT environment. The Helpdesk Technician provides first-level support to employees and handles common technical problems. When an issue involves network connectivity or network infrastructure, the Network Administrator can investigate and resolve the problem. The Linux System Administrator manages Linux-based servers and systems that support organizational services. The Cloud Administrator manages cloud-based resources and services when the organization uses cloud infrastructure. Through communication, documentation, escalation, and cooperation, the four roles help maintain reliable IT operations.

PART 7 – INFRASTRUCTURE RECOMMENDATIONS

7.1 Internet Provider

ABC Startup Solutions should subscribe to a reliable business-grade Internet service with sufficient bandwidth to support 20 employees, software development activities, cloud-based services, web browsing, file transfers, and other company operations.

A reliable Internet connection is important because the company will depend on network connectivity for communication, software development resources, online services, and access to external resources.

7.2 Server Specifications

A dedicated server should be provided for the company's centralized server requirements. The server should have sufficient processing power, memory, storage capacity, and network connectivity to support the company's initial requirements.

A proposed starting configuration is:

- Multi-core server processor
- 32 GB RAM

- 1 TB or larger storage capacity
- Gigabit Ethernet connectivity
- Redundant or reliable power protection
- Ubuntu Server LTS as the server operating system

The proposed specifications provide a starting point for a 20-employee organization while allowing room for future requirements.

7.3 Backup Strategy

ABC Startup Solutions should maintain regular backups of important company data. The company should use the NAS storage and external backup drives included in the infrastructure plan.

Important data should have more than one backup copy. Backup procedures should also be monitored regularly, and restoration should be tested to verify that backed-up data can be recovered when necessary.

7.4 Security Recommendations

The company should implement basic network and system security measures from the beginning because the scenario states that the company currently has no security policies.

Recommended measures include:

- Firewall protection
- Strong password requirements
- Regular software updates
- User access controls
- Antivirus/endpoint protection
- Regular backups
- Secure wireless network configuration
- Limited administrative access

- Regular monitoring of systems and network devices

These measures help reduce security risks and protect company systems and information.

7.5 Antivirus

Microsoft Defender should be used as the initial endpoint security solution for Windows computers because it is included with Windows and is already part of the required software inventory.

The antivirus solution should be kept updated and enabled on supported workstations.

7.6 Password Policy

ABC Startup Solutions should establish a password policy requiring employees to use strong and unique passwords.

The policy should include:

- Use of strong passwords
- Avoidance of easily guessed passwords
- Use of unique passwords for important accounts
- Protection of passwords from unauthorized disclosure
- Regular review of user accounts
- Appropriate access permissions based on employee responsibilities

7.7 Expansion Plan

The infrastructure should be designed with future expansion in mind. Additional workstations, network connections, storage, and other resources can be added as the company grows.

The network should maintain available capacity for additional devices, while the server and storage infrastructure should be reviewed and upgraded when the company's requirements increase.

Planning for expansion from the beginning can reduce the need for major infrastructure redesign when additional employees or resources are added.

PART 8 – PERSONAL REFLECTION

Working on this project helped me realize that system administration is not just about installing computers, setting up networks, or maintaining software. There is actually a lot of planning involved before anything can be installed or deployed. I learned that a system administrator needs to first understand what an organization needs and then decide what hardware, software, and network equipment would be appropriate. I also learned that proper documentation is important because it helps keep technical information organized and easier to understand.

The most challenging part for me was planning the network infrastructure and deciding what equipment would be needed for ABC Startup Solutions. Since the company was starting with no computers, server, network, Internet infrastructure, or security policies, I had to think carefully about how everything should be set up from the beginning. Creating the network diagram was also challenging because I needed to understand how the Internet, ISP modem, router, firewall, switch, server, printer, access point, and different departments would connect with one another.

I also realized that planning is very important before deploying an IT infrastructure. Good planning can help prevent unnecessary purchases, technical problems, and security issues. It also helps make sure that the available resources are used properly. By planning ahead, a company can have an infrastructure that meets its current needs while still allowing room for future growth. Without proper planning, the organization may experience problems with compatibility, limited resources, or difficulty adding new employees and equipment.

Overall, this project helped me improve my understanding of infrastructure planning, hardware and software inventory, network design, documentation, and decision-making. It also gave me a better idea of how different IT resources work together to support an organization. As a BSIT student, I believe this experience will be useful as I continue learning about system administration and prepare for future internship and work opportunities in the IT field.